

CS3550_C

THE UNIVERSITY OF WARWICK

Paper Details

Paper Code: CS3550_C

Paper Title: Digital Forensics

Exam Period: Summer 2023

Exam Rubric

Time allowed: 2 hours

Exam Type: Standard Examination

Approved Calculators: CASIO FX82, FX83 and FX85 series

Additional Stationary

Instructions

Answer any **FOUR** questions.

Read carefully the instructions on the answer booklet and make sure that the particulars required are entered on each answer booklet.

The exam question paper **MUST NOT** be removed from the examination venue.

1. (a) You have converted an RGB image to Y'CbCr color space. To reduce the color redundancy, you are about to perform subsampling on some color channel(s). Which color channel(s) should you work on for the subsampling?

[2 marks]

- (b) Given a certain color channel $\mathbf{I}$:

$$\mathbf{I} = \begin{bmatrix} 19 & 20 & 20 & 23 \\ 21 & 22 & 24 & 25 \\ 22 & 22 & 24 & 24 \\ 20 & 20 & 22 & 21 \end{bmatrix}$$

you are planning to subsample it in two different formats: a) 4 : 2 : 0 (left) and b) 4 : 1 : 1 (direct) formats. Show your subsampled results.

[6 marks]

- (c) It is popular to code image pixels in 8-bit, which has value ranges from [0, 255]. Assume the 4×4 matrix $\mathbf{I}$ (in 1. (b)) is an 8-bit grayscale image, how many bits are required to store it?

[2 marks]

In the fixed-length coding scheme, at least how many bits are required to code each entry in $\mathbf{I}$?

[3 marks]

In this case, show your codewords and calculate how many bits are required to store $\mathbf{I}$.

[5 marks]

- (d) There is an image with some salt and pepper noises (i.e., dead pixels) caused by faulty

sensors. Here is an example image patch: $\mathbf{J} = \begin{bmatrix} 19 & 22 & 18 \\ 21 & 255 & 21 \\ 19 & 24 & 22 \end{bmatrix}$.

You want to enhance the image quality by using some spatial filters (i.e., removing the dead pixels). Would you use mean filter or median filter? Calculate the filtered centre pixel (of $\mathbf{J}$) based on mean and median filters respectively, and justify your choice.

[7 marks]

2. (a) Assume we have a 4×4 grayscale image I (from 1. (b)). Answer the following questions:
- (i) Plot the image histogram, and the corresponding probability mass function (PMF). [4 marks]
 - (ii) What is the relationship between the histogram and PMF? [1 mark]
- (b) Gamma correction and histogram equalisation are two popular pixel-domain based contrast enhancement techniques. Can you describe them, including the major differences between the techniques? [8 marks]
- (c) In frequency domain, DC and AC components are two types of frequency coefficients. What is DC component? Assume you perform 2D-DCT on the 4×4 grayscale image I (from 1. (b)). Calculate the corresponding DC component. [5 marks]
- (d) As a forensic expert, you've managed to enhance a fingerprint image by removing the repetitive spectral noises, as shown in Fig. 1. Specifically, you've used Notch filters. In 100 words or less, explain the corresponding key steps for this image enhancement task. [7 marks]

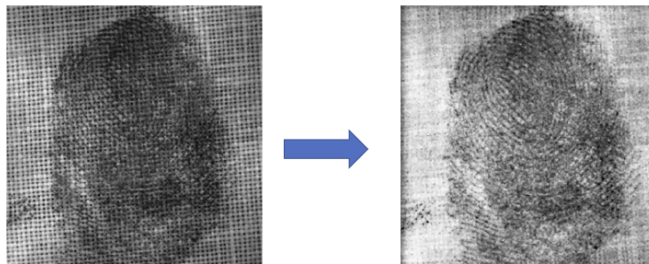


Figure 1: fingerprint enhancement by removing the repetitive spectral noises

3. (a) Assume you have two images of the same size: a binary image and an 8-bit grayscale image. Based on Least Significant Bit (LSB) substitution in the pixel domain, you've managed to embed the binary image as an invisible watermark to the grayscale image. Answer the following questions:
- (i) Why can LSB substitution yield invisible watermarks? [4 marks]
 - (ii) How can visible watermarks to be created by bit-plane substitution approaches? [2 marks]
 - (iii) List at least two approaches to destroy the invisible watermark (i.e., the one based on LSB substitution). [2 marks]
- (b) Watermark embedding can also be performed in the frequency domain, which is considered to be more robust than pixel-domain based approaches. Is it a good idea to embed watermark in high-frequency components? Justify your answer. [5 marks]
- (c) Given a watermark that is a random, Gaussian-distributed sequence, Spread Spectrum (SS) watermarking can be performed by embedding the watermark sequence to some DCT coefficients. At the decoding stage (i.e., to recover/extract the watermark), do you need the access to the original unwatermarked image? Justify your answer. [6 marks]
- (d) There are several popular (dis-)similarity metrics, such as Mean Squared Error (MSE), correlation coefficient, etc. yet image distortions may have some negative impact on these measures. Structural similarity (SSIM) index is a popular and effective solution in such noisy image assessment scenarios. In 80 words or less, given the reasons why SSIM is robust to image distortions. [6 marks]
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4. (a) JPEG compression can remove three main types of redundancies: spatial redundancy, psycho-visual redundancy, and coding redundancy. Explain how to remove these redundancies in details, i.e., operations and the related rationales. [9 marks]
- (b) Given a set of coefficients $\mathbf{c} = [21, 38, 17, 4, 23, 6]$ quantized by $q = 10$. Calculate the following:
- (i) quantized set of coefficients (use `round` operation during quantization); [2 marks]
 - (ii) de-quantized set of coefficients; [2 marks]
 - (iii) quantization error in terms of SSD (sum of squared difference). [2 marks]
- (c) Given a JPEG image, after de-compression you've found periodic empty bins in the histogram of DCT coefficients (e.g., with frequencies $u = 1, v = 1$ over all the 8×8 blocks). Does it mean this image has been tampered? [2 marks]
- (d) For a possibly spliced JPEG image, you have decided to use compression-based forensic approaches, e.g., to detect the JPEG ghosts. Specifically, this JPEG image is further compressed with respect to quality level (e.g., from 1 to 100), based on which quantization error (SSD) distribution is calculated and visualised. Answer the following questions:
- (i) Why does JPEG ghost use pixel values instead of DCT coefficients to compute the quantization error (SSD)? [3 marks]
 - (ii) The local minimum SSD values may indicate JPEG ghosts or false positives. How can we distinguish them from each other and are there any further operations that should be applied for this? [5 marks]
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5. (a) Circular shift&match is an exhaustive search approach to detect copy-move forgeries. During matching (between the original and the shifted images), there may exist false positives or false negatives. Explain how to use morphological techniques to process these two error types. [4 marks]
- (b) Block matching is another approach for copy-move forgery detection. Compared with exhaustive search, it is much faster. List the main steps of spatial-domain-based block matching, and explain which step makes it computationally efficient. [6 marks]
- (c) What is uniform Local Binary Pattern (LBP) code? How many uniform LBP codes are there for an 8-bit LBP coding scheme? Show the numbers separately for each case, and also list the LBP code with 1 transition. [7 marks]
- (d) Given multiple images taken by a particular camera, you are about to estimate the corresponding Sensor Pattern Noise (SPN) for source device identification. List the key steps for SPN estimation. [8 marks]
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